Exercise 2: Specification of Learning Tasks

(a) A cellar containing hundreds of wine bottles of different types.

Answer: O (Objects)
The wine bottles represent the real-world objects that the machine learning system will analyse.

(b) A table listing for each wine its measured values of acidity, sugar, alcohol, and pH.

Answer: X (Feature Space)
The table contains the feature values representing each wine sample, such as acidity, sugar, alcohol content, and pH.

(c) A professional sommelier who can accurately rate any wine as “high-quality” or “low-quality.”

Answer: γ(o) (True Classification Function)
The sommelier acts as an expert who assigns the correct quality label to each wine sample.

(d) A laboratory instrument that measures chemical features like acidity, sugar, and alcohol content.

Answer: α(o) (Feature Extraction Function)
The instrument measures the properties of real-world wine samples and converts them into numerical feature values.

(e) The set {HighQuality, LowQuality}.

Answer: C (Classes)
These are the two possible output categories into which the wine samples can be classified.

(f) The machine learning model you are trying to train that predicts wine quality from its measured features.

Answer: y(x) (Prediction Function / Classifier)
The trained machine learning model takes the extracted features as input and predicts whether the wine is of high or low quality.

Final Answer Summary

Item	Symbol	Meaning
(a)	O	Objects
(b)	X	Feature Space
(c)	γ(o)	True Classification Function
(d)	α(o)	Feature Extraction Function
(e)	C	Classes
(f)	y(x)	Prediction Function / Classifier